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Cite this: DOI: 10.1039/c9cp05808g

Unveiling the effect of electron tunneling on the plasmonic resonance of closely spaced gold particles†

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Recent experimental techniques enable the nanoparticles' (NPs) ensembles to be made with an angstrom-level of interparticle gap widths. The theoretical description of their optical properties becomes more challenging because of the nonnegligible quantum mechanism (QM) effects such as electron tunneling and nonlocal screening. To demonstrate the microscopic mechanism of QM effects and quantitatively elucidate their impact on a variety of surface plasmon resonance (SPR) models of gold particle oligomers, in this work we performed a theoretical study on closely spaced Au cluster dimers and NP oligomers by the first-principles approach, and the classical or quantum-corrected electromagnetic models (CEM or QCM), respectively. Through the first-principles calculation on a series of Au_N dimers with different interparticle distances d and N, we depicted the variation of the possibility of direct electron tunneling (DET) across the junction constructed by two nearest NPs with d and N, and found that it exactly follows an exponential decay and the decay rate linearly varies with $1/N$. The impact of the QM effect on the SPR models excited along the dimer axis is much more profound than those perpendicular to the dimer axis. CEM/QCM calculations on strongly coupled NP dimers and symmetric and asymmetric trimers demonstrated the evolution of their optical properties with variable NP sizes, gap separations and light polarization, as well as the QM effect on the major SPR modes. The side-by-side comparison between the results from time-dependent density functional theory and CEM/QCM models sheds light on understanding the origin of a variety of SPR models of gold NP oligomers and the QM effect on those modes, and makes a connection between the calculations of small cluster and large NP oligomers.

Received 25th October 2019,
Accepted 16th December 2019

DOI: 10.1039/c9cp05808g

rsc.li/pccp

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1 Introduction

Metallic NPs can assemble to form aggregates. In ensembles or aggregates, their individual plasmons couple together so that the near-field in the gaps can be enhanced by additional several orders of magnitude. The plasmonic enhancements significantly boost the efficiencies of nanoscale light–matter interactions and have enabled applications that include surface-enhanced Raman scattering for single-molecule detection,[1–3] optical nanoantennas,[4–6] high-harmonic generation,[7–9] solar energy harvesting,[10,11] and so on.[12] Coupling multiple nanoparticles together in chainlike structures has been suggested as an approach to nanoscale optical wave-guiding and focusing.[13–17] The tremendous progress in experimental techniques and fabrication of nano-plasmonic devices has stimulated theoretical development to address this complex physical phenomenon. Most of the existing theoretical studies employed either classical electromagnetic interaction[18–21] or the simplified plasmon hybridization or dipole–dipole coupling models[22–26] to describe the plasmon behaviors in the complex nanostructures of arbitrary shapes. These classical electromagnetic approaches provide a simple way to understand the plasmon coupling and are particularly well suited for weakly coupled metallic nanostructures.

Recent experimental techniques enabled the NPs' ensembles to be made with an angstrom-level of interparticle gap widths.[27–32] The experimental works have demonstrated that the quantum mechanical effects such as nonlocality[32–34] and electron tunneling[35,36] emerge as the gap-width approaches the atomic length-scale, which affect both the far-field radiation properties of plasmonic nanostructures and the near-field properties. In closely spaced NPs, the plasmon of individual particles can be

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† Electronic supplementary information (ESI) available: The experimental and fitting dielectric functions, an illustration of the QCM scheme, test of the layer number, test of the grid size dx, the absorption spectra, field enhancement and charge density distribution of additional NP dimer and trimer systems, the excitation energies and transition densities of the excited states of Au_N monomers and dimers with large oscillator strengths, and the atomic coordinates of all the Au_N clusters. See DOI: 10.1039/c9cp05808g

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significantly distorted due to the interparticle polarization effect and electron exchange interaction. The electron distribution in closely spaced NPs can be considerably different with that in weakly coupled NPs. As a result, the simple plasmon hybridization model fails, and theoretical models which can take into account these quantum effects, such as the full quantum model based on time-dependent density functional theory (TDDFT),[35,37–39] the nonlocal hydrodynamic (NLHD) model,[27,33,40,41] or the quantum corrected electromagnetic model (QCM),[36,42,43] are required. In fact these quantum models have predicted the spectral evolution of surface plasmon modes deviating from the classical electromagnetic model (CEM), and the reduction of near-field enhancement in an ultrasmall gap. Unfortunately, TDDFT calculations are computationally demanding even for small systems so that it cannot be used to describe typical plasmonic systems which usually contain millions or even billions of electrons. In the meantime, the NLHD model fails to describe the nanoparticle size effects on plasmon energies in free-electron materials such as alkali metals or aluminum, with its predicted blue-shift being contrary to the red-shift that are measured and obtained using the TDDFT calculations.[44,45] The QCM, which incorporates quantum-mechanical effects within a classical electrodynamic framework by introducing the fictitious medium in the gap of NPs, has been demonstrated to yield consistent results with TDDFT calculations.[36]

In this work, we performed a detailed theoretical study on the strongly coupled Au NP dimers and symmetric and asymmetric trimers with variable NP sizes, gap widths and the polarization of incident light by the QCM to address the effect of the electron tunneling across the junction in their optical responses. Both the optical absorption spectra and the near-field enhancement will be calculated, and the major plasmon resonance peaks will be characterized. A comparison with the results produced by the CEM, or to be more specific, the finite-difference time-domain (FDTD) approach, has been made. Here we seek an understanding on how the sizes and gap widths of NPs and the polarization direction dictate the excitation energy and absorption intensity of the plasmon resonance, and elucidate the impact of the DET effect on the plasmonic resonance of Au NP dimers and symmetric and asymmetric trimers.

Additionally, to understand the origin of DET across the junction formed by two nearest neighbouring nanoparticles, quantitatively characterize the decay rate of DET with the spatial separation and particle size, and describe its impact on the optical properties, we performed a first-principles (TD)DFT calculation on a series of small-sized Au cluster (Au_N) dimers. Our chosen clusters possess shapes of hollow cages and their cluster sizes fulfill the criterion of spherical aromaticity,[46] $N = 2(K+1)^2$ with $K = 2, 3, 4, 5$ (corresponding to Au_{18}, Au_{32},[47] Au_{50}[48] and Au_{72}[46]). These clusters have spherical shapes and have been proved to possess high thermodynamic stability.[46] Furthermore, it is computationally feasible to perform a many-body electronic structure calculation on these systems. Although there are other most stable structures like the core–shell particles[21,49,50] or the ligand protected clusters,[51–53] here we chose to study the neutral hollow cage clusters since they are free from the interference of ligand molecules and other atoms, the DET is only related to the property of pure gold clusters. By analyzing the variation of frontier MOs, absorption spectra and transition densities of the corresponding excited states with the gap widths and N, we give the physical consensus of how the DET is formed in a coupled NP pair and its impact on the optical responses. We will demonstrate that the capability of DET from one particle to another is pertinent to the coupling between the molecular orbitals of individual particles, and we will introduce a physical quantity to quantitatively characterize the variation of tunneling capability with the separation distance and NP size.

2 Theory and computational details

The classical electromagnetic simulations reported here are performed using the open-source program FDTD++ code, fully featured FDTD software, developed and maintained by Jeffrey M. McMahon.[54] The time-dependent Maxwell's equations are discretized using central-difference approximations with Yee's algorithm.[55] The light propagation direction is fixed along the $+z$ axis. The incident light pulse is injected from the plane at $z = 10.0$ nm with the functional form: $E(t) = e^{-\frac{(t-t_0)^2}{\sigma^2}} \sin(\omega_0 t)$, where $t_0 = 3.0$ fs, $\sigma = 0.1$ fs, and $\omega_0 = 3.10$ eV. The convolution perfectly matched layers are used to offer the best absorption in the box boundary. The dielectric function of gold is described by one Drude model term (conduction band electron motion), adding five Lorentz oscillators (interband electron transitions)[56] in the form of $\varepsilon(\omega) = \varepsilon_\infty - \frac{{\omega_D}^2}{\omega(\omega + i\gamma_D)} - \sum_{n=1}^{5} \frac{\Delta\varepsilon_{Ln}{\omega_{Ln}}^2}{\omega(\omega + 2i\delta_{Ln}) - {\omega_{Ln}}^2}$. All the parameters in dielectric function are determined according to the experimental data,[57] and the detailed fitting parameters are shown in Table 1. The fitting dielectric function closely matches the experimental result as depicted in Fig. S1 in the ESI.†

The QCM models the junction between adjacent nanoparticles by means of a local dielectric response considering electron tunnelling and tunnelling resistivity at the gap. The electron tunneling is mimicked by filling the junction with an artificial conductive material. A schematic illustration of the QCM treatment is shown in Fig. S2 (ESI†). The permittivity of this material is spatial dependent, and it is related to the distance between two separated NP surfaces as: $\varepsilon(l, \omega) = \varepsilon_\infty - \frac{{\omega_p}^2}{\omega^2 + i\omega\gamma_g(l)}$. The crucial tunneling damping γ_g is the key parameter describing the quantum tunnelling resistance introduced by the presence of the gap. In our QCM simulations, γ_g is assumed to follow the exponential relation of $\gamma_g = \gamma_D e^{ql}$, where $q = 2.24$ Å^{-1} is the effective growth rate of crucial

Table 1 The fitting parameters involved by the dielectric function

	Drude		Lorentz		
ε_∞	ω_D (eV)	γ_D (eV)	$\Delta\varepsilon_L$	ω_L (eV)	δ_L (eV)
3.00	8.8	0.072	0.40	2.76	0.30
			0.50	3.15	0.40
			1.84	4.15	1.00
			1.60	6.00	2.00
			−0.50	1.84	0.42

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tunneling damping with the separation l of the gold dimer[32,58] and γ_D is the damping parameter in our Drude term. The electromagnetic solutions are obtained by using FDTD++, which has been mentioned above. A cuboid simulation box with maximum 80 nm in length (along the dimer or trimer axis) and 60 nm in width and height is chosen. Along the dimer or trimer axis the grid size is set to be 0.1 nm while in other directions the grid size is set to be 0.2 nm. The influence of the grid size on the results has been tested. Fig. S3 in the ESI† shows that our chosen grid sizes are suitable.

3 Results and discussion

3.1 Au cluster dimers

To understand the capability of DET across the junction and depict on how the NP's size and spatial separation affect the tunneling probability, at first, we perform DFT/TDDFT calculations on a series of Au cluster dimers (see Fig. 1) within Gaussian 16 software.[59] The long-range-corrected DFT exchange–correlation functional ωB97X-D[60] will be adopted to calculated their ground- and excited-state properties. Four clusters, Au_{18}, Au_{32}, Au_{50} and Au_{72}, which possess nearly spherical shapes and whose average diameters are 0.54, 0.75, 0.96, and 1.18 nm, respectively, have been chosen as the model systems. By following our previous study on the Au cluster,[61] the geometrical optimization of Au_N is finished at the theoretical level of PBE/LanL2DZ with the deficient parameter set of Gaussian 16. Then the optimized clusters are directly used to construct Au cluster dimers with different spatial separation of d. No further geometric optimization is made for the dimers in order to maintain the spherical shape of Au_N. Here d denotes the distance between two nearest Au atoms in the gap. We set d to be $d \geq 2.8$ Å. 2.8 Å is the Au–Au bond length. In the latter FDTD simulation, no accurate atomic positions are known, and the corresponding spatial separation g of two NPs should approximately follow the relation of $g = d - 2.8$ Å.

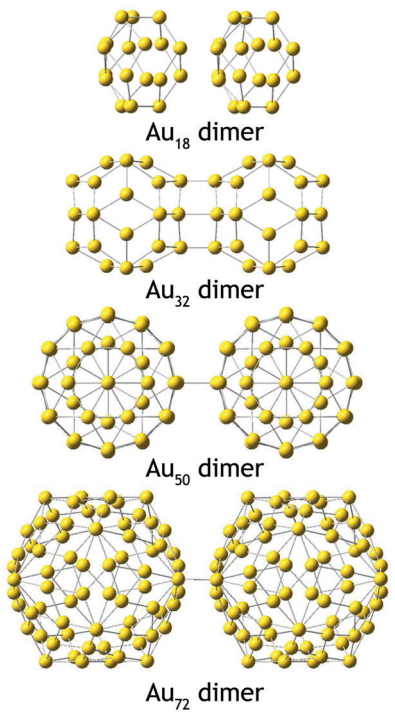


Fig. 1 The geometries of Au_N cluster dimers.

Fig. 2 and Fig. S4 in the ESI† show the variation of MO energy levels, the charge densities of the ground state and frontier MOs of Au_N pairs with d. All cluster pairs show very similar ground-state properties. For example, a considerable amount of charge is distributed in the gap of two cluster pairs, indicating the strong hybridization of two particles' wavefunctions, and the gap between the highest occupied MO (HOMO) and the lowest unoccupied MO (LUMO) becomes larger as d increases. The overlap between LUMOs of Au_N pairs at $d \leq 4$ Å is obvious, indicating the possibility of the electron transport or photo-induced interparticle charge transfer (CT). NP pairs with closely spatial separation may constitute a channel through which charge can flow freely. Contrary to the apparent overlap between LUMOs, the overlap between HOMOs is weak, indicating that the strength of the hybridization between the occupied molecular orbitals (OMOs) is appreciably weaker than that between unoccupied molecular orbitals (UMOs) since the OMOs are more localized than the unoccupied ones. The coupling between two NPs through OMOs (where the holes are excited) is way smaller than that of UMOs (where the electrons are excited), so we can deduce that the electrons can be more easily transported than the holes in Au NP dimers (see below).

To quantitatively describe the capability of DET across the junction from one site to the other, we define a quantity,

$$S_{AB}^{Occ/Uocc}(E_f) = \sum_{k}^{Occ/Uocc} \sum_{\alpha\in A} \sum_{\beta\in B} \frac{C_{\alpha k} S_{\alpha\beta} C_{\beta k}}{E_f - \varepsilon_k}, \quad (1)$$

which describes the propagation of a tunneling electron from cluster A to cluster B through the molecular orbitals.[62] Here E_f

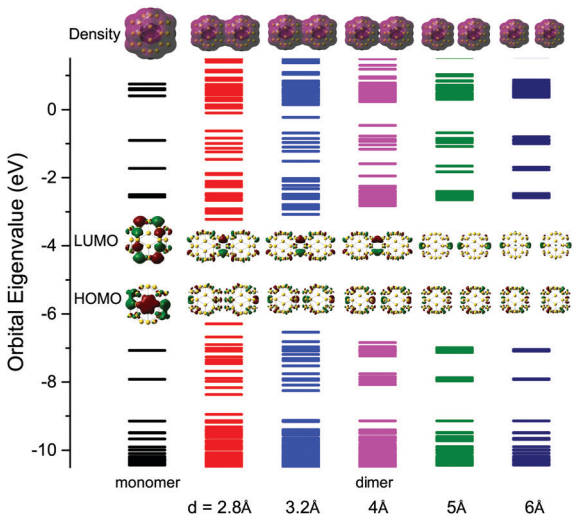


Fig. 2 MO energy levels, the charge densities of the ground state (on top), and frontier MOs (in the middle) of Au_{32} pairs varied with d.

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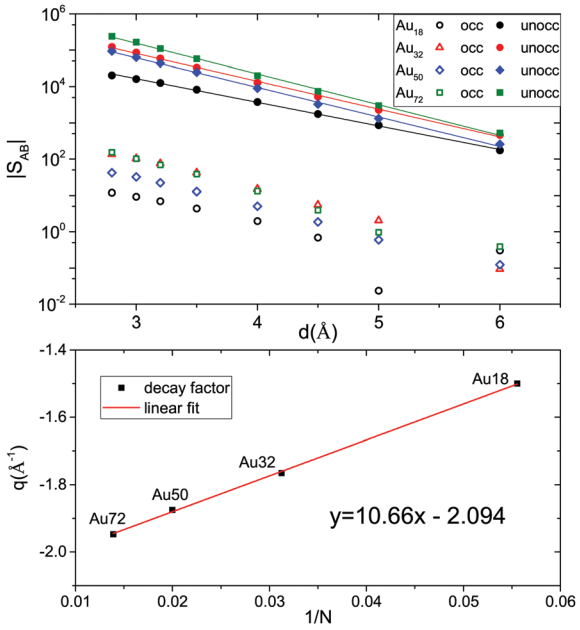


Fig. 3 Upper panel: $S_{AB}^{Occ/Nocc}$ varies with d for Au_{18}, Au_{32}, Au_{50}, and Au_{72} dimers. Lower panel: The linear fitting of decay rate q with $1/N$.

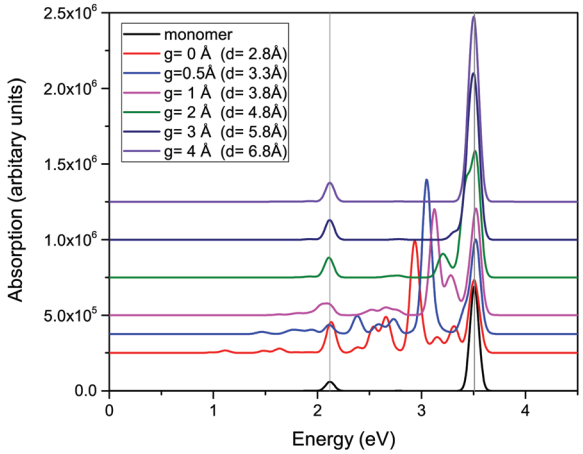


Fig. 4 Absorption spectra of the Au_{32} dimer varied with the spatial separation d from the TDDFT calculations.

is the Fermi energy which is located in the midgap of the HOMO and LUMO, ε_k is the k-th MO energy, C is the MO coefficient matrix, and S is the atomic orbital overlap integral matrix. We calculate $S_{AB}^{Occ/Uocc}$ for the Au_{18}, Au_{32}, Au_{50}, and Au_{72} dimers, and show the results in the upper panel of Fig. 3. It clearly demonstrates that S_{AB} decreases exponentially with the increase of d. This relationship between S_{AB} and d confirms that the DET across the junction can only play a role in the NP pairs with small spatial separation. At the same value of d, $|S_{AB}^{Uocc}|$ is three orders larger than $|S_{AB}^{Occ}|$, indicating that the DET in Au cluster pairs mainly takes places *via* the conduction bands. We fit the variation of $|S_{AB}^{Uocc}|$ with d as a function relation of e^{qd} to determine the decay rate q and find that q is inversely proportional to N as shown in the lower panel of Fig. 3. When $N \rightarrow \infty$, the intercept $q_0 = -2.094$ Å^{-1}. Interestingly, this value of $|q_0|$ is approximately equal to the exponential growth factor of γ_g[32,58] ($\gamma_g = \gamma_D e^{ql}$, $q = 2.24$ Å^{-1}). In the original QCM approach,[36] γ_g was assumed to be a function of the surface separation l of two NPs, and can be calculated from the static conductivity $\sigma_0(l) = \sigma(l,\omega \rightarrow 0)$ as $\gamma_g(l) = \frac{\omega_p^2}{4\pi\sigma_0(l)}$, where σ_0 can be obtained through quantum-transport theory or Green's function methods. Therefore, the exponential increase in γ_g with the spatial separation means that both the conductivity and tunneling probability decay exponentially with the spatial separation.

The DET and delocalized charge distribution definitely affect the optical responses of cluster pairs. Fig. 4 shows the calculated absorption spectra of the Au_{32} monomer and dimers with the different spatial separations. There are two major absorption bands in the spectrum of the Au_{32} monomer, located at around 2.1 and 3.6 eV, respectively. The low- and high-energy modes separately possess the molecular-like and plasmon-like characters verified by the transition density profiles shown in Table S1 in the ESI.† Considering the changes in spectral lineshapes of the dimers with reference to the spectrum of the Au_{32} monomer, the spectra of dimers can be explicitly divided into three categories. The first one shows no explicit change in spectral lineshape but only an overall spectral shift, and the second one shows an apparent spectral shift together with many additional new absorption modes appearing in the low-energy range. The first one takes place in the dimers with $d \geq 6.8$ Å and the second one takes place in the dimers with $d \leq 4$ Å, which is consistent with our previous observation.[61] Compared with the spectra of dimers with $d \geq 6.8$ Å, those of dimers with $d \sim 5$ Å show two major absorption bands around $\sim$3.4 eV, which is categorized into the third one.

In Table S1 in the ESI,† we listed the major characteristic modes which have relatively larger oscillator strengths, and we plotted their corresponding difference densities and transition densities (processed by the open source software Multiwfn[63]). When two clusters interact weakly, such as the dimers with d = 6.8 Å (g = 4 Å), their optical properties can be effectively explained in terms of the coupling of localized excitations in individual components of the nanostructures (A*B $\leftrightarrow$ AB*, where A*B/AB* means that the electrons and holes are placed on the same cluster), which is verified by the transition density profiles. For example for the Au_{32} dimer at d = 6.8 Å, its absorption band around 3.5 eV is composed of three major resonance plasmon modes, which are formed by the hybridization of two monomers' dipolar plasmon modes along three directions as shown in Table S1 in the ESI†. These modes are called the bonding dipolar plasmon modes (BDPs) later. In this case, the Frenkel exciton model is able to approximately explain the dimer's optical responses.

For more closely spaced dimers, the mutual polarization effect and electron exchange interactions between two clusters become more significant, as a result, the photo-induced intermolecular CT excitations take place and the CT excitation modes obviously mix with the localized plasmon excitations. For example, for the dimer with $d \sim 5.0$ Å, its BDP mode excited

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along the dimer axis direction is well separated from those along the perpendicular directions. The former usually possesses lower excitation energy than the latter two, ascribed to the mixture of localized plasmon excitations with intermolecular CT excitations. There are two main types of configuration interaction contributing to the excited dimers which weakly bound in the ground state and much more strongly interact in the excited electronic state. One is the exciton resonance (ER) caused by interaction between local excited states ($A^*B \leftrightarrow AB^*$), and the other is charge transfer resonance (CR) due to the interaction between CT states ($A^+B^- \leftrightarrow A^-B^+$). However, the transition density profiles only show the obvious mixture of ER and CR for the hybridized plasmon mode excited along the dimer axis, we thus deduce that the intermolecular CT excitations only take place along the dimer axis. In these cases, the spectra of the dimer can be properly explained by the mixed Frenkel-CT exciton modes.

When the junction-width is further reduced, such as the cases with d = 2.8 and 3.3 Å, two clusters are strongly interacted. Both their ground- and excited-state electronic wavefunctions strongly mixed together and the electrons are delocalized around the nanostructures and the junction, subsequently, there are many additional peaks in the absorption spectra. In these cases, the exciton model (both the Frenkel exciton and the mixed Frenkel-CT exciton models) fails to explain the optical responses of the dimers. The theoretical models which can account for the quantum tunneling and electron delocalization effects well are required.

3.2 Au NP dimers

Metallic particles in plasmonic applications are typically much larger than the above sized clusters. In the following, we thus calculate the Au NP systems with much larger sizes. Firstly, we focus on Au NP dimers, which are made of two identical Au nanospheres of different radii R and gap widths g. To have a close comparison on the results produced by CEM and QCM, we demonstrate the calculated absorption spectra of NP pairs of R = 5 nm by QCM in Fig. 5, and those by CEM in Fig. S5 (ESI†). In the following, unless explicitly stated, the incident electric field is assumed to polarize along the dimer axis.

QCM and CEM produce similar results for the dimers of $g > 2$ Å, where a dominant wide absorption band around 2.3–2.4 eV appears. The distinct difference appears at $g \leq 2$ Å, indicating that in this regime the quantum tunneling effect is significant, agreeing with our first-principles TD-DFT calculation on Au clusters. As g decreases from 2 to 0 Å, CEM predicts that the dominant broad absorption band gradually evolves into two well-separated absorption bands, and the one in the lower-energy region red shifts while the higher-energy one stays at around 2.39 eV with the decrease of g. From the plots of surface charge density distributions associated with two absorption bands as shown in the right panel of Fig. S5 (ESI†), we observe that this rapid red-shifted low-energy sharp peak possesses the character of the gap capacitance (GC) mode, and the high-energy one has the character of the BDP mode. As g decreases to sub-nanometer, more and more photo-induced charges are localized on the NP surfaces near the interparticle gap. As a result, the accumulated localized charges form a capacitance between the facing NP surfaces, which leads to the unphysical extreme strong field enhancement in the gap with the decrease of g as predicted by CEM.[25] When the DET effect gets involved by QCM, the changes in spectral lineshapes are dramatic. In the case of g = 1 Å, the absorption around 2.11 eV becomes weak and it becomes a shoulder of a major absorption band around 2.39 eV, which is the screened bonding dipole plasmon (SBDP) mode.[64] Additionally, a new absorption band appeared in the low-energy region of 1.06 eV, which comes from the photo-induced interparticle CT plasmon (CTP) excitation[65] as the corresponding charge density distribution

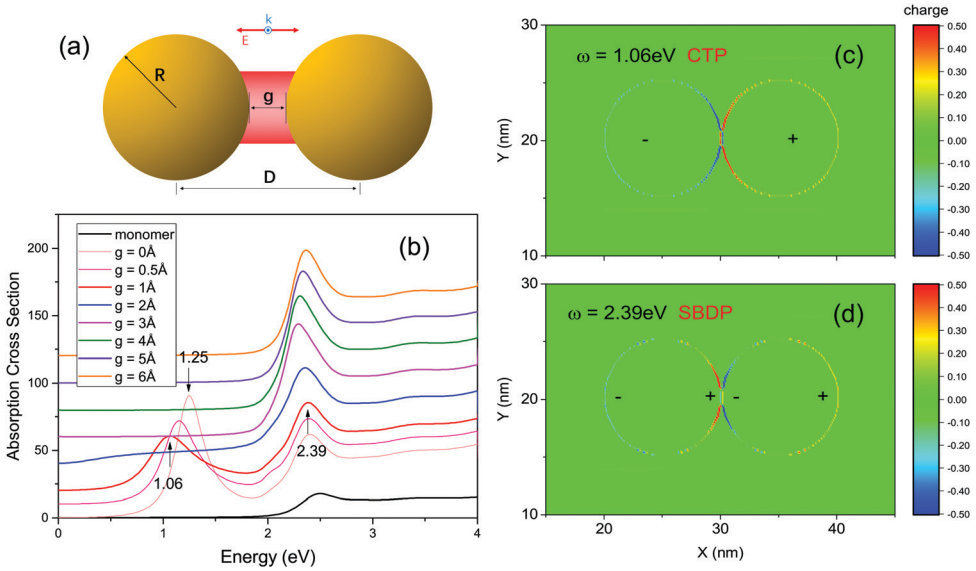


Fig. 5 (a) The illustration of the simulation model, where the electric polarization E is parallel to the dimer axis and the wave vector k is perpendicular to the paper plane. (b) The absorption spectra of a series of Au nanosphere dimers of R = 5 nm calculated by QCM. (c and d) The surface charge distribution of two selected modes for NP dimers with g = 1 Å.

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demonstrates. As g continually decreases, the CTP peak becomes stronger and sharper, blue-shifting in the meantime. The QCM results reveal that by getting the DET effect involved, the inter-particle CT excitation appears and the unphysical GC mode is eliminated. The variation of absorption spectra with g agrees well with our above TDDFT calculations of small clusters.

We have also calculated the absorption spectra of Au NP dimers of R = 7.5 and 10 nm to check the effect of NP sizes. The results are shown in Fig. S7 (ESI†). Compared with the results of R = 5 nm, the most distinct difference is that the increase of particle size can significantly enhance the absorption cross section, which can be explained by the variation of field-induced polarization. For a metal nanosphere, the polarizability is proportional to the volume of the NP ($P \propto R^3$). Therefore, the larger NP possesses a larger absorption cross section. Another difference is that the peak splitting between the GC and BDP modes becomes more obvious as R increases. The GC mode emerges before the DET effect starts to show as the gap separation narrows, so the peak splitting still exists for QCM results. In addition, we have plotted the variation of excitation energy and near-field enhancement associated with the strongest absorption peak (the CT mode is not included) with g and R in Fig. S8 in the ESI.†

Then we performed the calculation for the dimer with R = 5 nm and g = 1 Å by assuming different field polarization directions. ϕ is the angle between the light polarization and the dimer axis. Fig. 6 shows the evolution of spectra with ϕ. As ϕ increases from 0° to 90°, the major plasmon peak of the dimer starts with a 'red-shift' and ends with a 'blue-shift' compared with the position of the monomer's plasmon peak, and the major absorption bands of the dimers become broader than that of the monomer. Meanwhile, we observe that the impact of the electron tunneling effect on the spectra becomes less and less as ϕ increases, resulting in the reduction of the intensity of the CTP mode. The CTP mode fully disappears at ϕ = 90°, further indicating that the tunneling effect only affects the plasmon excitation along the dimer axis. The field enhancement and charge density distribution at ϕ = 45° are shown in Fig. S9 (ESI†). The characters of the corresponding major plasmon modes are similar to those in Fig. 5. However, here we observe the excitations both along and perpendicular to the dimer axis.

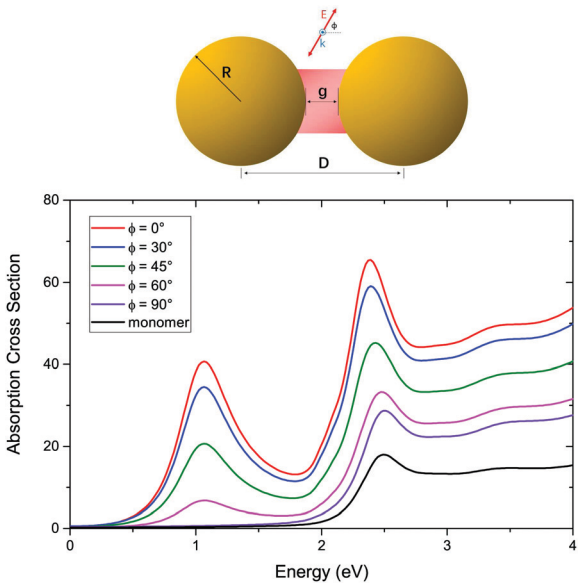


Fig. 6 The absorption spectra of Au dimers (R = 5 nm and g = 1 Å) varied with the light polarization direction.

3.3 Au NP trimers

Lastly, we performed a calculation for the Au NP trimers with the equal or unequal spatial separations (g_1 and g_2) between the two nearest neighbour nanoparticles. Fig. 7 shows the calculated absorption spectra of symmetric ($g_1 = g_2$) and asymmetric ($g_1 \neq g_2$) trimers. For the symmetric trimers, the evolution of spectra with the increase of g is similar to that of the dimer. The major absorption band firstly red shifts and then blue shifts as $g_1(g_2)$ decreases from 6 to 1 Å. When the gap separation is small, the CTP absorption band also emerges at a low energy region. However, the peak position is red-shifted compared to the dimer cases (1.06 → 0.89 eV for g = 1 Å, 1.25 → 1.07 eV for g = 0 Å). This is caused by the coupling between CTP modes formed among NPs, leading to the formation of a bonding charge transfer (BCT) mode and an antibonding charge transfer (ACT) mode. The latter one is a dark state in the symmetric situation $g_1 = g_2$, because the opposite transition dipole moments are canceled. The apparent difference between the results of the symmetric trimer and dimer is that the absorption cross section of the

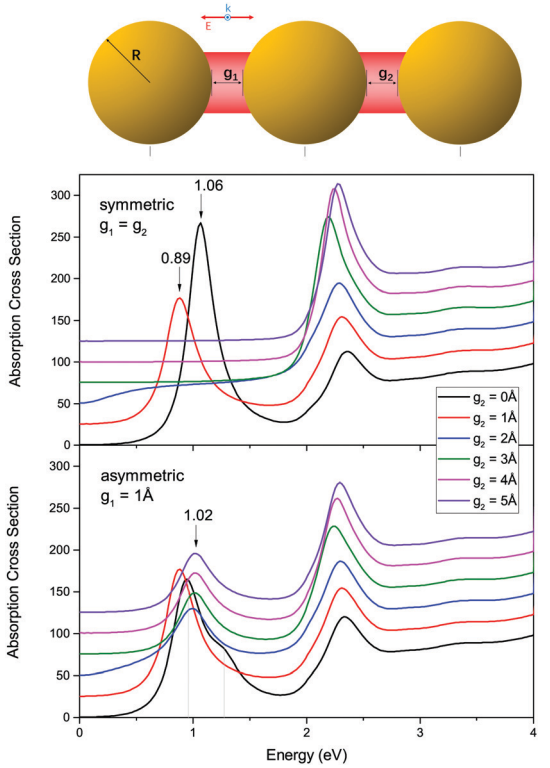


Fig. 7 The calculated absorption spectra of symmetric and asymmetric trimers with R = 5 nm by QCM. Upper panel: The symmetric trimers with variable gaps of g_1 and g_2. Lower panel: The asymmetric trimers with variable g_2. g_1 is set to be 1 Å.

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trimer is much larger than that of the dimer, and the location of the plasmon peak of the trimer red shifts more rapidly than that of the dimer in the same gap separation. This is because the mutual electromagnetic coupling among trimers is much larger than that among dimers. These findings are expected to play an important role in designing plasmonic nanostructures for practical applications that require coupled metallic NPs with superior properties. One can increase the number of nanoparticles in the chain to enhance the absorption cross sections and tune the locations of plasmon resonance modes.

The lower panel of Fig. 7 shows the absorption spectra of the asymmetric trimer. Here $g_1 \neq g_2$, and g_1 is fixed to be 1 Å and g_2 is variable. With this arrangement of nanostructures, there always exists the lower-energy CTP band in absorption spectra no matter how large g_2 is, since in this asymmetric structure, g_1 is in the quantum tunneling region and the electrons can always tunnel between the left two NPs. The influence of the third NP also makes the CTP mode red-shift a little (1.06 → 1.02 eV) compared to the g = 1 Å dimer situation. On the other hand, at g_1 = 1 Å and g_2 = 0 Å, the spectrum shows two separated CTP peaks. However, the peak positions are different in the case of g = 1 Å and 0 Å (1.06 → 0.95 eV, 1.25 → 1.35 eV). This can be explained by the hybridization of two CTP modes, forming a lower energy BCT mode and a higher energy ACT mode. Due to the broken symmetry ($g_1 \neq g_2$), now the ACT mode has a relatively small yet non-zero transition dipole moment. Thus there are two CTP peaks, one red-shifted strong BCT mode and one blue-shifted weak ACT mode in the spectrum, which is verified by the plots of charge distributions of two CT modes shown in Fig. 8.

The factors of near-field enhancement in the gap centers of symmetric and asymmetric trimers are calculated and plotted in Fig. S10 in the ESI.† The symmetric trimer ($g_1 = g_2$) has a larger field enhancement factor than the asymmetric one (g_1 = 1 Å), indicating that the electron tunneling taking place in the junction formed by NP 1 and 2 affects the charge density distribution of the third one, so that the factor of field enhancement in the center of gap 2 is always smaller than that in the symmetric trimer. In other words, the local electric fields formed in adjacent two gaps are coupled together.

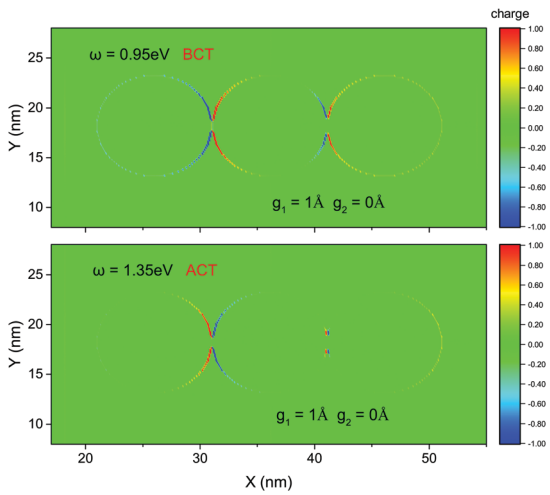


Fig. 8 Local surface charge distributions for two CT modes of the symmetric trimer at g_1 = 1 Å and g_2 = 0 Å calculated by QCM.

4 Concluding remarks

We have performed a theoretical study on the optical responses of closely spaced spherical Au particles by the TD-DFT and quantum-corrected electrodynamic approach to elucidate the impact of electron tunneling across the junction on their optical properties. The sizes of particles vary from clusters to nanoparticles, and the strongly coupled Au NP dimers and symmetric and asymmetric trimers with variable sizes, spatial separations and light polarization have been investigated. Both the absorption spectra and the near field enhancements in the gaps have been calculated, and the major plasmon modes have been characterized. Additionally, a (TD-)DFT calculation on a series of Au_N dimers has been performed to understand the microscopic mechanism of DET across the junction, to depict its decay rate with the increasing spatial separation and cluster size, and to characterize their impacts on the optical properties of Au_N dimers. A side-by-side comparison between the results from TDDFT and CEM/QCM models has been made. In the following, we briefly summarize the main findings of the present work:

(1) The capability of DET across the junction formed by two Au clusters decreases exponentially with the spatial separation of d so that DET only obviously affects the optical responses of closely spaced particles with the gap separation of $d \leq 4.8$ Å ($g \leq 2$ Å). The DET rate is also sensitive to the cluster's size. As N increases, the magnitude of decay rate $|q|$ increases and $|q_0| = 2.094$ Å^{-1} when $N \to \infty$. Interestingly, this value is very close to the exponential growth factor of γ_g, which is adopted to characterize the artificial conductive materials filled in the junction of NPs.

(2) The TD-DFT calculation on the small clusters reveals that the interparticle CT excitations mainly couple with the hybridized dipolar plasmon mode excited along the dimer axis. Subsequently, this coupling leads to the separation of plasmon modes excited along and perpendicular to the dimer axis so that we observe two peaks appearing around 3.5 eV for the dimers with $d \sim 5$ Å and one peak for the dimers with $d \geq 6.8$ Å.

(3) The QCM calculations for larger NP dimers with varied polarization directions of incident field confirm the above observation. The GC modes usually lie below the major hybridized dipolar plasmon modes and appear in the spectrum when the NP size is large enough. The low-energy CTP modes which are well separated from the major plasmon modes only show up in the optical spectra of NPs with $g \leq 2$ Å and $\phi \neq 90^\circ$.

(4) The NP trimer shows a stronger absorption cross section and near-field enhancement than the dimer with the same g and R. In the asymmetric trimer, the electron tunneling appearing between NP 1 and 2 affects the local field of gap 2 formed by NP 2 and 3. The asymmetric trimer has a broader absorption band than the symmetric one. At smaller separations, we observe the coupling between two CTP modes and the coupling among CTPs and localized plasmon excitations as well.

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This work sheds light on understanding the origin of electron tunneling through the junction and its impact on the optical properties of nanostructures, and guiding the rational design of nano-structures tailored for specific applications such as surface-enhanced spectroscopies where one seeks to maximize the electromagnetic field enhancement, or energy harvesting where efficient absorption of light is necessary.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

Financial support from the National Natural Science Foundation of China (Grant No. 21573177 and 21833006) is acknowledged.

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